

Hamiltonian-Geometric Optimization

Speaker Script and Anticipated Q&A (12-Slide Deck)

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This document accompanies `hamiltonian_geometric_short_presentation.tex`. Part 1 is a read-aloud script, one block per slide (each roughly 30–45 seconds, so the full talk runs about 7–9 minutes). Part 2 is a set of hard, likely audience questions with grounded answers, meant to be read *after* the talk, in preparation for questions — not to be presented.

1 Speaker Script

Slide 1 — Title

“Today I’m presenting the Hamiltonian-geometric optimization framework — a way of designing optimizers by treating training itself as a physical system. This is a short walkthrough: the math, how we validated it, and what the benchmarks actually show.”

Slide 2 — Core Question

“The question we’re testing is simple to state: can one Hamiltonian-geometric update rule work across very different settings — classical ML, LLM-style training, physics-informed networks, and quantum control? We treat the model’s parameters as coordinates, the loss as potential energy, and the optimizer’s momentum as a physical momentum moving on a curved metric manifold. And I want to be upfront about the claim: it’s bounded. We are not claiming this beats Adam everywhere — only that it’s strongest when the problem actually has exploitable curvature.”

Slide 3 — Methodology Flowchart

“This is the whole pipeline in one picture, so you don’t have to hold every equation in your head at once. We start from the parameters and the loss, build a metric from the regularized Hessian, plug that into a Hamiltonian, and get the equations of motion out of it. Three correction forces — geometric, memory, and spectral-entropy — feed into one generalized update rule, which also collapses back down to familiar optimizers like SGD or Adam as special cases. Everything below this box is validated first, analytically, before we ever look at a benchmark curve.”

Slide 4 — Hamiltonian Formulation

“Here’s the actual Hamiltonian: kinetic term is momentum contracted through the inverse metric, potential term is the loss. Differentiating gives the equations of motion, and the interesting part is the second term in the momentum equation — that’s the geometric force, and it only exists because the metric depends on where you are in parameter space. We add dissipation on top, because plain

Hamiltonian mechanics conserves energy, and we want the loss to actually go down, not oscillate forever.”

Slide 5 — One Update Rule, Many Optimizers

“The point of this slide is that we didn’t invent a totally separate optimizer family. SGD and heavy-ball fall out when the metric is flat. Adam and RMSProp fall out when you keep only the diagonal of the metric. Natural gradient is the full metric without the momentum and memory terms. Our Hamiltonian-geometric optimizer is what you get when you keep everything. And modern optimizers like Shampoo, K-FAC, or Muon all sit somewhere on that same spectrum, from a full matrix down to a single scalar.”

Slide 6 — Analytical Validation

“This slide answers the question people always ask: how do you know these equations are actually correct? Every equation here traces back to a specific reference: my own literature-review PDF, which is the design document for this optimizer’s architecture. We didn’t just implement it and hope — we wrote a separate numerical validation module, `paper_math_validation.py`, that checks each equation with finite differences on a fixed, seeded toy problem where the metric genuinely curves with the parameters. Three kinds of checks: ‘literal’ means we tested an equation exactly as the PDF prints it. ‘Completed’ means the PDF states something but never actually derives it — like the gradient of the spectral entropy — so we derived a specific closed form ourselves and checked it against a numerical gradient. ‘Counterexample’ is the honest one: the PDF’s condition for the Hessian regularization doesn’t actually guarantee a positive-definite metric, we show that failure on purpose, and then confirm our shipped code fix repairs it, over hundreds of random trials. All seven checks pass to numerical precision.”

Slide 7 — Benchmark Evidence Snapshot

“Four results at once here, on purpose, so you can see the workload variety in one glance: MNIST accuracy, an AlgoPerf-style validation loss curve, DeepOBS-style validation accuracy, and an industry-style LLM run on WikiText-2. Across these, the Hamiltonian-geometric optimizer posts the best or near-best validation loss and accuracy. But these are small-scale runs — I’d call this evidence of correct mechanism, not a scaling proof. The next slide puts exact numbers on all of this.”

Slide 8 — Benchmark Results Table (Classical, PINN & LLM)

“So here are the actual numbers behind those curves. On MNIST, AlgoPerf-style, DeepOBS-style, and the capacitor PINN, Hamiltonian-geometric reports the best validation or test loss of everything we compared against. The LLM row is the one I want to be honest about: on WikiText-2, AdaFactor actually edges us out slightly on validation loss — 6.765 versus our 6.778. But we still clearly beat AdamW and Lion, and we finish fastest of the five in wall-clock time. Muon-lite diverges entirely here, which is a reminder that these are untuned, single-seed runs, not a scaling claim.”

Slide 9 — Modern Optimizer Comparison

“Now the harder comparison: against modern optimizers people actually use today. Left panel is compute cost versus accuracy — Adam, our method, and Muon lead the pack. Middle panel is a 10-seed sweep, and the near-tie holds up under that noise. Right panel is where it gets more honest: as we scale model width, Adam scales a bit more consistently than we do. So: competitive, genuinely,

but not a decisive win. And to be transparent, the version benchmarked here is a diagonal-metric reduction, not the full curvature-memory-entropy architecture.”

Slide 10 — Benchmark Results Table (Modern Optimizers)

“And the numbers behind that comparison. Ranked by best-epoch validation loss on the spiral MLP: Adam, Hamiltonian-geometric, and Muon are within four ten-thousandths of each other — that’s noise, not a real ordering. AdamW and Lion trail by a clearer margin, and Shampoo is both slower per epoch and slightly worse here. The honest read is a narrow leading group of three, not a win for any one of them.”

Slide 11 — Limitations

“Four honest caveats. Scale: everything here is small; a full n -by- n metric is a mechanism demonstration, not something we’re claiming replaces Adam in production. Statistics: most results are single-seed; the 10-seed sweep is the only place we have real variance data. Tuning: learning rates come from modest grids, not exhaustive search. And comparability: the modern-optimizer benchmark uses a reduced, diagonal version of our method, so that comparison specifically isn’t testing the full framework.”

Slide 12 — Summary and Thank You

“Three takeaways. The formulation holds together once you’re explicit about the spectral-force sign and the memory-metric construction. The empirical evidence is strongest exactly where the theory predicts it should be — where there’s real, exploitable curvature. And against modern optimizers, we’re in the leading group, not decisively ahead of Adam or Muon. Happy to take questions, especially on the validation methodology or where we’d want to push this next.”

2 Anticipated Questions and Answers

The questions below are ordered roughly hardest-and-most-important first. Each answer is grounded in a specific number or result already in the codebase (the numeric validation module and the consolidated report), so they can be defended on the spot rather than improvised.

1. **How do you actually know these equations are correct, rather than just plausible-looking physics?**

We don't validate against the benchmark tasks themselves — we validate the equations directly, symbol by symbol, in `src/paper_math_validation.py`. Each of the seven checks compares a closed-form expression from the reference PDF against an independent finite-difference computation on a fixed, seeded synthetic problem, to machine precision (errors of 10^{-9} to 10^{-16}). Critically, this toy problem was built specifically so the metric $g(\theta)$ depends on θ ; the project's own capacitor benchmark has a metric that is *constant* in θ , which would make F_{geo} and $\nabla_{\theta} S(g)$ trivially zero and prove nothing.

2. **Is the improvement over Adam statistically significant, or just noise?**

Borderline, and we say so explicitly. On the 10-seed modern-optimizer sweep, a paired t -test (licensed because every optimizer trains on the same paired θ_0 /dataset draw per seed) gives HG vs. Adam: $t(9)=-2.234$, $p=0.052$ — borderline, HG lower on 7 of 10 seeds. HG vs. Muon: $t=-0.838$, $p=0.424$ (not significant). HG vs. AdamW: $t=-1.457$, $p=0.179$ (not significant). $n = 10$ is underpowered to settle this either way; the honest claim is a weak directional signal against Adam specifically, and no signal against Muon or AdamW.

3. **The memory force is supposed to help, so why does it hurt on the quantum benchmark?**

On the unsafeguarded kicked-top ablation, turning on the memory force alone moves final loss from 0.0491 (the $\beta=0.7$ default) to 0.0631 — worse. The explicit geometric force is worse still ($0.0491 \rightarrow 0.1949$) and over $10\times$ slower, because both differentiate a metric that is itself built from finite differences. Only the spectral-entropy force helps on this task ($0.0491 \rightarrow 0.0184$). The honest reading: momentum tuning dominates on this benchmark, and the three correction terms have a *mixed*, task-dependent effect, not a uniformly positive one. Separately, on the PINN ablation, the picture flips: under a toy (parameter-independent) metric only the memory term is significant ($p=7.4 \times 10^{-4}$, a small $\sim 12\%$ effect), but under the true Hessian metric the spectral term becomes highly significant ($p=3.2 \times 10^{-8}$) while memory and geometric forces are not. The takeaway we lead with is that these terms matter most — and differently — depending on whether the metric is actually tied to real loss curvature.

4. **On WikiText-2, AdaFactor beat you. Doesn't that undercut the "works across domains" claim?**

Yes, on that specific metric, and we report it rather than hide it: AdaFactor's validation loss is 6.765 against our 6.778. But it's one 200-step, single-seed, untuned run — not a scaling claim. In the same run we clearly beat AdamW (6.843) and Lion (7.032), and finish fastest in wall-clock time; Muon-lite diverges outright (loss 9.31), which itself shows how sensitive these small, untuned comparisons are. The claim we defend is "competitive across domains," not "best in every domain."

5. **Isn't " $\lambda > 0$ guarantees positive-definiteness" obviously true? Why call that a bug?**

It’s a real counterexample, not pedantry: for the indefinite Hessian $\text{diag}(-5, 0.1, 3)$ and the PDF’s literal $\lambda=10^{-3}$, the regularized matrix $H + \lambda I$ still has a negative eigenvalue, because λ is far smaller than the most negative eigenvalue’s magnitude. A fixed constant can never work for every Hessian; the regularization has to depend on the matrix’s own spectrum. The project’s shipped fix, `src.pinn._positive_definite_metric`, does exactly that, and we verify it produces a strictly positive-definite metric on that counterexample and across 200 random indefinite trials.

6. You add a dissipation term, but Hamiltonian systems conserve energy. Isn’t that a contradiction?

It’s intentional, not a contradiction we’ve glossed over: pure Hamiltonian flow conserves H exactly, which is useless for optimization — we need the loss to decrease. The Rayleigh dissipation term is deliberately added on top of the conservative dynamics, making the overall system a damped (port-Hamiltonian-style) system rather than an energy-conserving one. The "Hamiltonian" in the name describes where the structure (metric, momentum, curvature) comes from, not a claim that the implemented dynamics conserve energy.

7. The modern-optimizer comparison uses a diagonal reduction of your method — isn’t that just comparing Adam to Adam?

Largely, yes, and the report says this outright: by our own special-case correspondence, that diagonal reduction is already "a diagonal-metric approximation to the Hamiltonian optimizer," i.e. close in spirit to Adam with a different momentum discretization. The near-tie there is expected, not evidence of the framework’s distinctive machinery beating Adam. The full machinery — F_{geo} , F_{mem} , the spectral term together — is only exercised on the PINN (exact Hessian metric) and quantum-control benchmarks.

8. Could this scale to a billion-parameter model?

Not as implemented, and we don’t claim it could. A full $n \times n$ metric costs $O(n^2)$ memory and roughly cubic cost to invert or factor, which is infeasible past small networks — exactly why Shampoo and K-FAC exist, using Kronecker-factored approximations to cut that cost (e.g. $O((mn)^3) \rightarrow O(m^3 + n^3)$ for a layer shaped $m \times n$). Getting a cheap structured approximation of our full metric, the way Shampoo/K-FAC do, is future work, not something we’ve benchmarked.

9. How is this actually different from just using natural gradient, K-FAC, or Shampoo?

Those methods are different points on the *same* metric-structure hierarchy we use to classify optimizers — specific approximations to the same underlying Hamiltonian object (full metric $\rightarrow$ Kronecker-factored $\rightarrow$ diagonal $\rightarrow$ scalar). What none of them include is our other two ingredients: the non-Markovian memory force built from an exponentially-weighted gradient history, and the spectral-entropy regularization on the metric’s own eigenvalue distribution. The claim is that curvature alone (natural gradient/K-FAC/Shampoo) is one piece; momentum-as-phase-space-dynamics and the two correction forces are what’s distinctive here.

10. What result would actually change your mind about this framework?

Two things would: if, under a genuine full-metric setting (the exact-Hessian PINN or quantum-control benchmarks) with many more seeds, the additive correction terms showed no significant, reproducible effect at all — rather than the mixed, sign-flipping effect we currently see — that would undercut the core mechanistic claim. And if no cheap structured approximation to the full metric could be found, that would cap the framework’s practical relevance at small-scale, mechanism-level demonstrations rather than anything deployable.